Fundamentals of Probability: Set Theory Basics

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Sets and Subsets

Definition: A set is a collection of distinct objects. A subset is a set whose elements are all contained in another set.

- Example: If $A = \{1, 2, 3\}$, then $B = \{2, 3\}$ is a subset of A.
- Notation: $B \subseteq A$ (subset), $B \neq A$ (proper subset)

Power Set: The set of all subsets of a given set.

- Example: If $A = \{1, 2\}$, then $P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

Set Operations

Basic Operations:

- **Union:** $A \cup B = \{x | x \in A \text{ or } x \in B\}$
- **Intersection:** $A \cap B = \{x | x \in A \text{ and } x \in B\}$
- **Difference:** $A - B = \{x | x \in A \text{ and } x \notin B\}$
- **Complement:** $A^c = \{x | x \notin A\}$

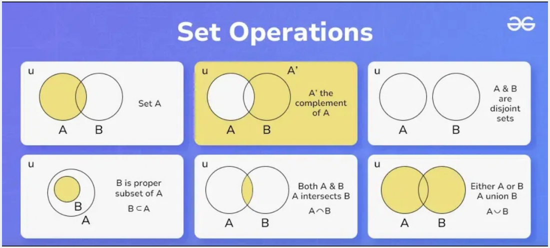
Example: If $A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$:

- $A \cup B = \{1, 2, 3, 4\}$
- $A \cap B = \{2, 3\}$
- $A - B = \{1\}$, $B - A = \{4\}$

Venn Diagram Representation

Visualizing Set Operations:

- Union: Combine all elements
- Intersection: Common elements
- Difference: Elements in one set but not the other



Counting Rules

Multiplication Rule:
If a process involves two or more independent steps, and if the first step can occur in n_1 ways, the second step in n_2 ways, and so on, then the total number of outcomes is the product of the number of ways each step can occur. For two steps, this is given by:

$$n_1 \times n_2.$$

More generally, for a sequence of k steps where the ith step can occur in n_i ways, the total number of outcomes is:

$$\prod_{i=1}^{k} n_i.$$

Counting Rules Contn'd...

Addition Rule:
If there are two or more mutually exclusive events (i.e., events that cannot occur simultaneously) such that one event can occur in n_1 ways and another in n_2 ways, then the total number of outcomes is the sum of the ways in which each event can occur:

$$n_1 + n_2.$$

More generally, for mutually exclusive events $E_1, E_2, \ldots, E_k$ that occur in $n_1, n_2, \ldots, n_k$ ways respectively, the total number of outcomes is:

$$\sum_{i=1}^{k} n_i.$$

Counting Rules Contn'd...

Multiplication Rule Example

Suppose you want to create a two-character password. The first character is an uppercase letter (with 26 possibilities) and the second character is a digit (with 10 possibilities). Since each choice is independent, the multiplication rule tells us that the total number of possible passwords is:

$$26 \times 10 = 260.$$

Addition Rule Example

Imagine you have two separate fruit baskets. One basket contains 3 apples and the other contains 4 oranges. If you are to pick exactly one piece of fruit, you can either pick an apple or an orange, but not both at the same time (i.e., the events are mutually exclusive). Therefore, by the addition rule, the total number of ways to pick one fruit is:

$$3 + 4 = 7.$$

Counting Rules Contn'd...

Permutations (Ordered Arrangements) Number of ways to arrange r objects from n:

$$P(n,r) = \frac{n!}{(n-r)!}$$

Example: Arranging 3 books from 5:

$$P(5,3) = \frac{5!}{(5-3)!} = 60$$

Counting Rules Contn'd...

Combinations (Unordered Selections) Selecting r objects from n:

$$C(n,r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

Example: Selecting 2 students from A, B, C.

- Permutations count order, giving 6 ways: (AB, AC, BA, BC, CA, CB)
- Since AB and BA are the same selection, we divide by 2!:

$$\frac{3!}{2!(3-2)!} = \frac{6}{2} = 3$$

- Valid selections: {AB, AC, BC}.

Generalization:

- Permutation formula overcounts by $r!$ due to order.
- Divide by $r!$ to remove redundancy.
- Thus, we get:

$$C(n,r) = \frac{n!}{r!(n-r)!}$$

Counting Rules Contn'd...

Examples for Permutations and Combinations
Distinct Objects:

- **Permutation:** Assigning 3 students to 3 different seats in a row from a group of 10.
  Solution: $P(10,3) = \frac{10!}{(10-3)!} = 720.$
- **Combination:** Choosing 3 students from 10 to form a committee.
  Solution: $C(10,3) = \frac{10!}{3!(10-3)!} = 120.$

Examples for Repetition Allowed Cases
Non-Distinct Objects:

- **Permutation with Repetition:** Creating a 3-digit password from digits (0-9).
  Solution: $10^3 = 1000.$
- **Combination with Repetition:** Choosing 3 scoops of ice cream from 5 flavors.
  Solution: $C(5+3-1,3) = C(7,3) = 35.$

Binomial Theorem Expanding $(a+b)^n$:

$$(a+b)^n = \sum_{r=0}^{n} \binom{n}{r} a^{n-r} b^r$$

Example: Expansion of $(x+y)^4$:

$$x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

Rule 5: Inclusion-Exclusion Principle Formula:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Example: In a class of 50 students:

- 30 like Python
- 25 like Java
- 10 like both

Solution: Students liking at least one:

$$30 + 25 - 10 = 45$$

Fundamentals of Probability: Counting Techniques

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Probability Space

A **probability space** is a triple $(\Omega, \mathcal{F}, P)$ consisting of:

- A sample space,Ω, which contains all possible outcomes of the experiment.
- A set of events $\mathcal{F}$ which must be a σ-algebra.
- A Probability measure P that assigns probabilities to the events in $\mathcal{F}$

σ-Algebra

A σ-algebra $\mathcal{F}$ is a collection of sets in Ω such that:

1. If a set $S \in \mathcal{F}$, then $S^c \in \mathcal{F}$.
2. If the sets $S_1, S_2 \in \mathcal{F}$, then $S_1 \cup S_2 \in \mathcal{F}$. This also holds for infinite sequences; if $S_1, S_2, \dots \in \mathcal{F}$, then $\bigcup_{i=1}^{\infty} S_i \in \mathcal{F}$.
3. $\Omega \in \mathcal{F}$.

Probability Measure

A probability measure is a function defined over the sets in a σ-algebra $\mathcal{F}$ such that:

1. $P(S) \geq 0$ for any event $S \in \mathcal{F}$.
2. If the sets $S_1, S_2, \ldots, S_n \in \mathcal{F}$ are disjoint (i.e., $S_i \cap S_j = \emptyset$ for $i \neq j$), then
$$P\left(\bigcup_{i=1}^{n} S_i\right) = \sum_{i=1}^{n} P(S_i).$$
   Similarly, for a countably infinite sequence of disjoint sets $S_1, S_2, \cdots \in \mathcal{F}$,
$$P\left(\lim_{n\to\infty} \bigcup_{i=1}^{n} S_i\right) = \lim_{n\to\infty} \sum_{i=1}^{n} P(S_i).$$
3. $P(\Omega) = 1$.

Measure

A **measure** is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying:

- **Non-negativity:** For all $A \in \mathcal{F}$, $\mu(A) \geq 0$.
- **Null empty set:** $\mu(\emptyset) = 0$.
- **Countable additivity (σ-additivity):** If $\{A_i\}_{i=1}^{\infty}$ is a countable collection of disjoint sets in $\mathcal{F}$, then:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

Examples of Measures 1. Lebesgue Measure: Assigns length, area, or volume to subsets of $\mathbb{R}^n$.e.g., The Lebesgue measure μ on $\mathbb{R}$ assigns to an interval $I = [a, b]$ its length:

$$\mu([a, b]) = b - a$$

Examples of Measures Contn'd...

Example: Consider the disjoint intervals:

$$A = [1, 2] \quad \text{and} \quad B = [3, 5]$$

The Lebesgue measure of $A \cup B$ is:

$$\mu(A \cup B) = \mu(A) + \mu(B) = (2 - 1) + (5 - 3) = 1 + 2 = 3$$

2. Counting Measure: Assigns to each set the number of elements it contains.

Conditional Probability

Definition: The probability of event A occurring given that event B has already occurred is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad \text{if } P(B) > 0.$$

Key Points:

- Used when events are **dependent** (one event affects another).
- If A and B are **independent**, then:

$$P(A \mid B) = P(A).$$

Chain Rule

Let $(\Omega, \mathcal{F}, P)$ be a probability space and $S_1, S_2, \ldots$ a collection of events in $\mathcal{F}$. Then,

$$P\Big(\bigcap_i S_i\Big) = P(S_1)\, P(S_2 \mid S_1)\, P(S_3 \mid S_1 \cap S_2) \cdots$$

or equivalently,

$$P\Big(\bigcap_i S_i\Big) = \prod_{i=1}^{\infty} P\Big(S_i \,\Big|\, \bigcap_{j=1}^{i-1} S_j\Big).$$

Law of Total Probability

Definition: Let $(\Omega, \mathcal{F}, P)$ be a probability space and let the collection of disjoint sets$B_1, B_2, ... \in \mathcal{F}$ be any partition of Ω. For any set $A \in \mathcal{F}$

$$P(A) = \sum_i P(A \mid B_i) P(B_i).$$

Example: Student Pass Probability

- 40% of students take **math** (B_1), 60% take **science** (B_2).
- The probability of passing is 80% in math and 50% in science.

Calculation:

$$P(A) = P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2)$$

$$= (0.8 \times 0.4) + (0.5 \times 0.6) = 0.32 + 0.30 = 0.62.$$

Bayes' Theorem

Definition: A way to find the probability of a **cause** given that an **effect** has occurred.

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

- $P(H)$: Prior Probability (initial belief about hypothesis H)
- $P(E|H)$: Likelihood (probability of evidence E given H)
- $P(E)$: Total Probability of Evidence
- $P(H|E)$: Posterior Probability (updated belief after observing E)

Bayes' Theorem

By using Law of Total probabilty, Bayes'theorem takes the following form:

Formula:

$$P(B_k \mid A) = \frac{P(A \mid B_k)P(B_k)}{\sum_i P(A \mid B_i)P(B_i)}.$$

Example: Medical Testing

- A test is **90% accurate** if a person has a disease.
- A false positive occurs **5% of the time**.
- The disease occurs in **1% of the population**.

Calculation:

$$P(Disease \mid Positive) = \frac{(0.9 \times 0.01)}{(0.9 \times 0.01) + (0.05 \times 0.99)}$$

$$= \frac{0.009}{0.009 + 0.0495} = \frac{0.009}{0.0585} \approx 0.154.$$

Even with a positive test, there is only a **15.4% chance** the person has the disease.

Intuition Behind Bayesian Probability

1. Start with a **prior belief(prior probability)** about an event.
2. Collect **evidence** (new data).
3. Update the belief using **Bayes' theorem** give **posterior probability**.

Example in Machine Learning Spam Detection:

- Prior: Probability an email is spam, $P(\text{spam})$.
- Likelihood: Probability of seeing certain words given spam, $P(\text{features}|\text{spam})$.
- Posterior: Updated probability that the email is spam given its content.

Used in: Naïve Bayes classifier, Bayesian Networks.

Summary of Differences

- **Conditional Probability:** Finding $P(A)$ knowing B has occurred.
- **Law of Total Probability:** Computing $P(A)$ by considering all possible ways it can happen.
- **Bayes' Theorem:** Finding $P(B \mid A)$ (cause from effect) using known conditional probabilities.

Comparison Table:

Concept	**Formula**
Conditional Probability	$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$
Law of Total Probability	$P(A) = \sum P(A \mid B_i)P(B_i)$
Bayes' Theorem	$P(B_k \mid A) = \dfrac{P(A \mid B_k)P(B_k)}{\sum P(A \mid B_i)P(B_i)}$

Summary

- Probability is the study of uncertainty and measures how likely an event is to occur.
- It is used in various fields such as statistics, finance, science, and AI.

Basic Definitions

- **Experiment:** Any process that leads to well-defined outcomes.
- **Random Experiment:** An experiment where the outcome cannot be predicted with certainty before it occurs. e.g., Tossing a coin
- **Sample Space** (S): The set of all possible outcomes.
- **Event** (E): A subset of the sample space.
- **Probability of an Event** $(P(E))$: A number between 0 and 1 representing likelihood.

Types of Events in Probability

In probability, an **event** is a subset of the sample space S. Here are common types of events:

- **Sure Event:** Always occurs, $P(A) = 1$.
- **Impossible Event:** Never occurs, $P(A) = 0$.
- **Simple Event:** An event with only one outcome (e.g., rolling a 4 on a die).
- **Compound Event:** An event with multiple possible outcomes (e.g., rolling an even number).
- **Mutually Exclusive Events:** Events that cannot occur together.
- **Independent Events:** The occurrence of one does not affect the other.
- **Complementary Events:** The event A^c where $P(A) + P(A^c) = 1$.
- **Equally Likely Events:** Events that have the same probability of occurring. $P(A) = P(B) \quad$ for all outcomes in S Example: Rolling a fair die (each face has $P = \frac{1}{6}$).

Types of Probability

- **Classical Probability**: $P(E) = \frac{\text{Favorable outcomes}}{\text{Total outcomes}}$
- **Empirical Probability**: Based on experimental data:

$$P(E) = \frac{\text{Occurrences of } E}{\text{Total trials}}$$

- **Subjective Probability**: Based on personal judgment.
- **Axiomatic Probability (Kolmogorov)**
  - Non-negativity: $0 \leq P(E) \leq 1$.
  - Certainty: $P(S) = 1$.
  - Additivity: If E_1 and E_2 are disjoint (mutually exclusive),

$$P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

Fundamental Laws of Probability

The basic probability laws are:

- **Non-negativity Law:** $0 \leq P(A) \leq 1$ for any event A.
- **Additive Law (For Mutually Exclusive Events/Disjoint Sets):**

$$P(A \cup B) = P(A) + P(B), \quad \text{if } A \cap B = \emptyset.$$

- **General Addition Rule:**

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

- **Multiplication Law (For Independent Events):**

$$P(A \cap B) = P(A)P(B).$$

- **Complement Law:**

$$P(A^c) = 1 - P(A).$$

- **Conditional Probability:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}, \quad \text{if } P(B) \neq 0.$$